

World Models for Cognitive Agents: Transforming Edge Intelligence in Future Networks

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ABSTRACT

World models are emerging as a transformative paradigm in artificial intelligence, enabling agents to construct internal representations of their environments for predictive reasoning, planning, and decision-making. By learning latent dynamics, world models provide a sample-efficient framework that is especially valuable in data-constrained or safety-critical scenarios. In this article, we present a comprehensive overview of world models, highlighting their architecture, training paradigms, and applications across prediction, generation, planning, and causal reasoning. We compare and distinguish world models from related concepts such as digital twins, the metaverse, and foundation models, clarifying their unique role as embedded cognitive engines for autonomous agents. We further propose Wireless Dreamer, a novel world model-based reinforcement learning framework tailored for wireless edge intelligence optimization, particularly in low-altitude wireless networks (LAWNs). Through a weather-aware UAV trajectory planning case study, we demonstrate the effectiveness of our framework in improving learning efficiency and decision quality.

INTRODUCTION

In the science fiction film *The Matrix*, the eponymous system serves as a platform that forecasts future outcomes based on the protagonist's decisions. This fictional technology presents a compelling vision: if artificial intelligence (AI) models can accurately internalize the workings of the real world, they could unlock the ability to perform complex tasks through predictive reasoning. This idea is now gaining significant attention in AI research: *how can AI, like humans, learn the fundamental laws governing our world and make informed predictions?*

Motivated by this idea, world models have emerged as a class of AI systems that learn an internal model of the environment's dynamics, such as its physical laws and spatial properties.¹ By

predicting future observations and rewards, they provide an agent with an "imagination" space for various tasks, including planning and control. Concretely, world models are typically built upon generative AI (GenAI) frameworks trained to reconstruct past observations and forecast future trajectories. By implicitly capturing patterns of the environment, the world model estimates what is likely to happen next. Rather than relying solely on immediate sensor data, the agent uses this internal model as a forward-looking guide, much like a chess player thinking several moves ahead or a driver intuitively expecting a pedestrian to jaywalk. In essence, the world model grants the AI agent a cognitive capacity to perceive, anticipate, and reason about its environment, enabling it to make more informed and adaptive decisions under uncertainty.

While the world model acts as the brain that provides the agent with an understanding of the real world, its specific focus varies depending on the task. For generation tasks, world models act as imagination engines, creating plausible future scenarios and generating remarkably realistic video sequences that adhere to physical laws, emphasizing that these models capture fundamental principles [1]. In terms of planning, such as autonomous driving, a world model allows an agent to mentally simulate action sequences and predict unexpected events, enhancing its robustness in real-world execution [2]. When it comes to reasoning, the agent leverages world models to anticipate the consequences of its decisions, infer hidden causal factors, and support what-if analysis for more informed decision-making [3].

Given these strengths, world models are increasingly recognized as foundational for advanced cognitive agents. As humans use mental models to understand and navigate their environment, AI systems embed world models at the core of their cognitive architecture to guide both perception and action. Many state-of-the-art systems, such as Wayve's GAIA model for safer

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assisted and autonomous driving,² exemplify this approach. Looking ahead to next-generation networks, where countless smart devices and autonomous systems operate at the edge, world models can be transformative. Edge agents, from self-driving cars and delivery drones to intelligent sensors, must often make instant decisions locally, without relying on constant cloud connectivity. By equipping them with robust world models, these agents can predict, plan, and act directly on-site, improving responsiveness and reducing dependence on cloud back-ends. Companies such as NIO have already embraced this paradigm, developing the NIO World Model specifically for on-car AI predictions.³

Building on these foundations, this article provides a comprehensive perspective on world models, covering from concept to deployment. We then introduce several applications and highlight their potential impact in the realm of edge intelligence. Furthermore, we propose a world model framework, Wireless Dreamer, for wireless edge intelligence optimizations. Unlike traditional optimization or reinforcement learning (RL) methods, our approach leverages a world model to recover and predict data for more effective decision-making in low-altitude wireless networks (LAWNs). As an early study, this article demonstrates how world model-based reinforcement learning can be realized and applied to wireless edge intelligence scenarios in LAWNs.

The key contributions of this work are summarized as follows:

- We investigate the concept of world models and clarify their role in wireless edge intelligence by analyzing fundamental challenges such as partial observability, time-varying environments, and long-term decision coupling, and by distinguishing world models from related paradigms.
- We propose Wireless Dreamer, a world-model-based framework for wireless edge intelligence optimization, where latent wireless dynamics are learned to enable predictive perception and foresighted decision-making.
- We study a weather-aware UAV-assisted communication task in low-altitude wireless networks as a case study, and demonstrate that Wireless Dreamer accelerates convergence and improves decision quality compared with representative baselines.

FROM EDGE CHALLENGES TO WORLD MODELS

CHALLENGES IN EDGE INTELLIGENCE

Sequential decision-making in highly time-varying environments is a fundamental challenge in realizing edge intelligence. Unlike traditional static networks, advanced edge systems, such as LAWNs enabled by UAVs, tightly integrate communication with autonomous control in scenarios where network states evolve continuously and unpredictably. In such scenarios, intelligent agents must optimize communication performance metrics, including throughput and coverage, while simultaneously operating under rapid temporal variations in physical environments [4]. In summary, the key challenges in developing edge intelligence for such dynamic decision-making include:

- **Highly Time-Varying Wireless Dynamics:**

Wireless edge agents operate in non-stationary environments where channel characteristics shift rapidly over time. Factors such as fast-fading signals, intermittent interference, and the continuous mobility of UAVs create a landscape where the optimal policy is never static. Achieving robust intelligence requires the agent to adapt to these volatile fluctuations in real-time.

- **Temporal Coupling of Decisions:** Agents' actions are coupled over time, where current decisions, such as battery consumption for mobility or increased transmit power, directly shape future network capabilities [2]. Effective decision-making, therefore, requires foresight to account for long-term performance and sustainability, rather than focusing solely on the immediate state.
- **Partial Observability:** Despite rich sensing, agents only observe incomplete and noisy snapshots of the environment, while many critical factors, such as latent interference sources, user behavior, or hidden mobility patterns, remain unobservable [4]. Intelligent operation thus requires inferring latent dynamics from historical observations and reasoning over unobserved states.

Taken together, these challenges reveal that edge intelligence is not merely a reactive control problem, but a fundamentally predictive decision-making task. An intelligent agent must reason beyond instantaneous observations and anticipate the long-term consequences of its actions under partial observability.

EVOLUTION FROM MODEL-FREE TO PREDICTIVE AGENTS

To address the shift from reactive control to predictive decision-making, the prevailing paradigm in wireless research has relied on model-free RL, such as Deep Q-Networks (DQN) [5]. While effective in static settings, these agents map instantaneous observations directly to actions without explicitly modeling the environment. This limits their applicability in the edge scenarios discussed above. Particularly, model-free agents adapt slowly to non-stationary wireless conditions and lack the ability to anticipate the long-term consequences of temporally coupled decisions [4].

To bridge the gap between reactive learning and predictive intelligence, several advanced paradigms have been explored. Although closely related, these paradigms address different objectives and operate at different levels of abstraction:

- **Digital Twins:** Digital twins provide high-fidelity virtual replicas that continuously mirror the state of their real-world counterparts for system monitoring and analysis [6]. Through simulation, they can further support predictive assessment and forecasting to assist agent decision-making. However, they are typically realized as externally deployed, cloud-based systems, rather than as internal models residing within the agent itself.
- **The Metaverse:** The metaverse refers to a persistent virtual environment that integrates physical and digital spaces, supporting real-time interaction and simulation. It offers a flexible platform for training, evaluation, and exploration of future scenarios during the training phase of edge intelligent systems [7]. Never-

Based on these functional categories, the world model integrates knowledge learning and environment modeling to create a comprehensive system capable of complex interactions with dynamic environments, supporting various downstream tasks.

¹ <https://www.nvidia.com/en-sg/glossary/world-models>

² <https://wayve.ai/thinking/gaia-2/>

³ <https://www.nio.cn/smart-technology/20241120002>

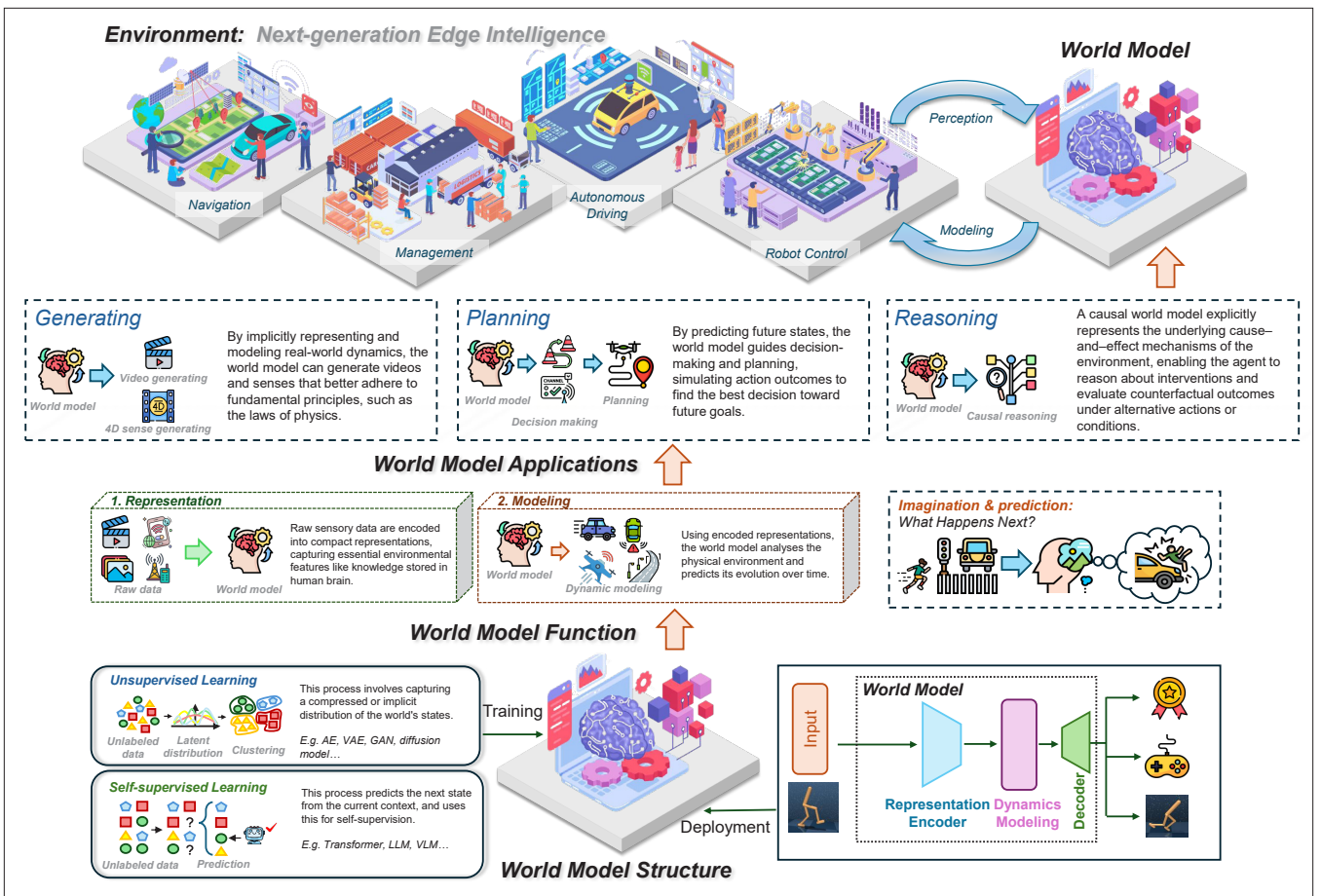


FIGURE 1. Illustration of world models for next-generation edge intelligence. The bottom part shows the workflow and structure of world models, including their training and deployment. The middle part illustrates the core functions of world models. The top part highlights the three core applications of world models and their potential application in a real-world environment.

theless, it is mainly positioned as an enabling infrastructure for experimentation and collaboration, rather than being directly used for autonomous decision-making at runtime.

- **Foundation Models:** Foundation models capture broad and transferable knowledge across diverse tasks [8]. While highly expressive, they may not explicitly encode the fast-varying, localized dynamics of wireless environments, such as rapid channel fluctuations or topology changes. For edge intelligence, it is often more critical to incorporate environment-specific dynamics that facilitate efficient prediction and planning [4].

Consequently, there is a critical need for an intrinsic cognitive architecture that internalizes environmental dynamics to enable foresight-driven planning directly at the edge. This necessity drives the paradigm shift towards **world models**, which equip agents with a latent “imagination” space to simulate future trajectories and optimize decisions proactively.

GENERAL CONCEPT OF WORLD MODELS

A world model is an internal representation to simulate and predict environmental dynamics. It serves as a cognitive map or simulator, encoding the external world’s state in compressed latent variables and modeling its temporal evolution, allowing the agent to anticipate outcomes and plan actions accordingly [3]. World models can be broadly classified into two functional categories:

- **Implicit Representation:** These models capture environmental dynamics as latent representations, supporting more informed decision-making. The transformation of real-world phenomena into implicit knowledge enhances decision-making by enabling the models to represent and reason about complex world knowledge.
- **Dynamics Modeling:** These models use current observations to predict future events by modeling continuous spatial and temporal dynamics. For instance, large vision models (LVMs) and vision-language models (VLMs) [8], such as Sora and Kling AI, can generate video content that to some extent captures physical regularities observed in real-world dynamics, including motion trajectories and object interactions.

Based on these functional categories, the world model integrates knowledge learning and environment modeling to create a comprehensive system capable of complex interactions with dynamic environments, supporting various downstream tasks. As depicted in Fig. 1, the general workflow and structure of world models includes:

- **Encoder:** First, the raw sensory inputs are processed and encoded into a structured internal representation. Through dimensionality reduction and feature extraction, these diverse and complex observations are transformed into compact and meaningful representations that reflect the essential characteristics of the environment.

World Model Comparison with Other Models					
Name	Definition	Function	Technical Foundation	Applications	Advantages
World Model [3]	An internal representation that an intelligent system uses to understand and predict environmental dynamics	- Imagination - Representation - Dynamics modeling - Support for downstream tasks	- GenAI - Unsupervised/self-supervised learning - Internet of things sensing	- Robot navigation - Autonomous driving - Game playing	- Future state prediction - Reasoning and decision-making - Generalization & adaptability
Digital Twin [6]	A high-fidelity virtual replica of a physical system, designed to mirror the state of its real-world counterpart in real time	- Condition monitoring - Performance analysis - Predictive maintenance - Remote control	- Internet of things sensing - Real-time data streaming - Physics-based simulation - Edge & cloud computing	- Smart city utilities - Aerial Omniverse Twin - City-Scale Spectrum Twin	- Risk-free testing - Visualization & monitoring - Cost-saving predictive maintenance
Metaverse [7]	A persistent, interconnected virtual world blending physical and digital realities, where users and AI interact in real time	- Social interaction - User-generated content - Economic transactions - Immersive experiences	- Physical devices 5G/6G network - Blockchain & digital assets - AI-driven content generation	- Immersive social platforms - Digital exhibitions - Virtual events	- Immersive interaction - Open innovation ecosystem - Physical-digital fusion - Training & validation platform
Foundation Models [8]	A broad-coverage neural network pre-trained on massive, heterogeneous corpora to distill general representations for adaptation	- Universal representation learning - Few-/zero-shot adaptation - Multimodal fusion and cross-modal reasoning	- Unsupervised/self-supervised learning - Alignment techniques - Mixture-of-experts & retrieval-augmented modules	- Text, code, image, and audio generation - Autonomous robotics and embodied AI - Scientific discovery	- Broad knowledge base spanning many domains - Emergent reasoning and planning abilities - Rapid task transfer
World Model Downstream Applications					
Downstream Tasks	Characteristics	Applications	Architecture	Paper & Code	
Generating	- Prioritizes long-horizon predictability - Embeds physical constraints, e.g., conservation laws, dynamics equations - Each frame is a dynamically consistent rollout from previous states	- Long-form video synthesis - Action-conditioned video generation 4D reconstruction	- WorldDreamer - Adapting video diffusion (AVID)	- Wang, Xiaofeng, et al. "Worlddreamer: Towards general world models for video generation via predicting masked tokens." arXiv preprint arXiv:2401.09985 (2024). Available code at: world-dreamer.github.io - Rigter, Marc, et al. "Avid: Adapting video diffusion models to world models." arXiv preprint arXiv:2410.12822 (2024). Available code at: sites.google.com/view/avid-world-model-adapters/home	
Planning	- Provides an explicit multi-step rollout of future trajectories - Elevates policy evaluation based on the full future distribution - Widens the decision horizon, easing sparse- or delayed-reward problems	- Game playing - Long-task control - Autonomous driving	- DreamerV2 - UniZero	- Hafner, Danijar, et al. "Mastering atari with discrete world models." arXiv preprint arXiv:2010.02193 (2020). Available code at: github.com/danijar/dreamerv2 - Pu, Yuan, et al. "Unizero: Generalized and efficient planning with scalable latent world models." arXiv preprint arXiv:2406.10667 (2024). Available code at: github.com/pendlib/LightZero	
Reasoning	- Models the full action-system-outcome causal chain - Predicts "what-if" ripple effects to assess system-wide consequences - Enables rich counterfactual analysis and interpretable inference	- Legal and forensic causality analysis - Causal reasoning	- Reasoning via Planning (RAP) - Large World Model (LWM)	- Hao, Shibo, et al. "Reasoning with language model is planning with world model." arXiv preprint arXiv:2305.14992 (2023). Available code at: github.com/Ber666/llm-reasoners - Liu, Hao, et al. "World model on million-length video and language with blockwise ringattention." arXiv preprint arXiv:2402.08268 (2024). Available code at: largeworldmodel.github.io/lwm	

TABLE 1. Comparison of world models with other models and their downstream applications.

- **Modeling:** Utilizing encoded representations, the world model effectively models and analyzes the physical environment. Subsequently, the model predicts the evolution of the physical environment over time. Accurate dynamics modeling allows for imagining future states, enabling proactive and anticipatory responses from agents.

- **Decoder:** By leveraging predictive understanding of future states, the model ultimately supports diverse downstream tasks such as autonomous driving and robot control, through decoding its implicit representations.

It is worth noting that the last part, the decoder, is a downstream task-dependent module, so the same world model can be adapted to different downstream tasks by swapping or fine-tuning different decoders [9].

DEPLOYMENT OF WORLD MODELS

The deployment of world models relies on GenAI techniques to internalize environmental data. This process typically employs two complementary paradigms that align with the model's core functions:

- **Unsupervised Learning (Representation):** To process high-dimensional sensory inputs such as radar signals and camera images, world models utilize unsupervised learning to compress raw data into compact latent states. Generative frameworks, such as Variational Autoencoders (VAEs) and diffusion models, are used here to capture the underlying spatial structure and static features of the environment without requiring manual labels [10]. This establishes the agent's internal Implicit Representation.
- **Self-supervised Learning (Prediction):** To master environmental dynamics, world mod-

els utilize self-supervised learning, where the model predicts the next latent state based on the current context. By treating the future sequence of states as the supervision signal, architectures including Transformers learn the temporal evolution of the system [11]. This enables the prediction capability necessary for foresight and planning.

In summary, unsupervised learning constructs the state space understanding "what currently is," while self-supervised learning constructs the transition dynamics predicting "what happens next," as shown in Fig. 1.

COMPARISON WITH OTHER MODELS

To strictly delineate the world model from other paradigms, we focus on its unique role as an embedded cognitive engine. Unlike digital twins, which aim for high-fidelity external mirroring, a world model prioritizes an internal, abstract representation optimized for decision efficiency rather than perfect accuracy. Unlike the metaverse, which serves as an external training platform, the world model acts as the runtime inference core for autonomous control. Finally, in contrast to the broad generalization of foundation models, world models focus on compressing specific environmental dynamics. In summary, for edge devices that perform autonomous decision-making under limited bandwidth and computational resources, the world model represents the most suitable endogenous modeling paradigm. A comprehensive comparison of these paradigms is summarized in Table 1.

APPLICATIONS OF WORLD MODELS

Beyond their conceptual formulation, world models have begun to demonstrate practical value in real-

Unlike conventional video generation networks, a world model prioritizes long-horizon predictability and approximate consistency with underlying physical dynamics.

world systems. In autonomous driving, they allow agents to anticipate pedestrian movements and traffic dynamics, improving safety and planning efficiency [2]. Wireless edge networks have the potential to enable UAVs to dynamically adjust their trajectories based on real-time environmental conditions.

PREDICTING

The core application of the world model is to learn implicit expression and prediction. Formally, a world model is trained to approximate the latent Markov process, thereby internalizing how the environment evolves from one instant to the next. For example, in the seminal work [12], a recurrent network was used as a world model to predict the next state and reward based on the agent's action in the CarRacing game. Through ablation studies, the authors demonstrated that removing this world model led to a 30% drop in gaming score and a significant increase in variance [12]. Unlike traditional RL, which directly maps observations to actions, a world model-based approach leverages predictive modeling to anticipate outcomes. These predictive capabilities improve action selection and support downstream tasks, such as control, forecasting, and anomaly detection.

GENERATING

For generative tasks, the central objective is to produce coherent temporal data, video streams, or 4-D scene reconstructions. Unlike conventional video generation networks, a world model prioritizes long-horizon predictability and approximate consistency with underlying physical dynamics. For example, WorldDreamer is a video world model that learns universal motion and physical dynamics, which are embedded in visual signals [13]. Built on a Spatial-Temporal Patchwise Transformer, the model restricts attention to localized windows in both space and time, allowing it to capture fine-grained visual dynamics efficiently. Remarkably, it can synthesize a high-quality 24-frame video at 192×320 resolution in just 3 seconds on a single A800 GPU [13].

PLANNING

For planning tasks, the chief objective is to synthesize anticipatory action sequences that maximize long-term returns. Unlike conventional Markov-decision processes (MDPs) or RL, where each policy update relies solely on the current state, a world model furnishes the planner with a prediction of future contingencies, effectively widening the decision horizon beyond one-step transition probabilities. For instance, DreamerV2 learns long-horizon behaviour entirely within this learned world model, employing an actor-critic framework to learn the policy [9]. The actor selects actions by imagining sequences of compact latent states, while the critic aggregates the predicted future rewards to capture returns that lie beyond the planning horizon. Notably, DreamerV2 attains human-level performance on the Atari benchmark, achieving a gamer-normalized median score above 2.0 compared with the human gamer baseline of 1.0 [9].

REASONING

In reasoning tasks, the primary goal is to reveal the underlying causal mechanisms that drive the environment's behavior. A world model simu-

lates and anticipates how actions will ripple through the system. For example, the authors in [14] proposed a LLM reasoning framework called Reasoning via Planning (RAP). RAP constructs a reasoning tree guided by a world model to identify high-reward reasoning paths that properly balance exploration and exploitation. In various challenging reasoning tasks, RAP combined with LLaMA-33B even outperformed chain-of-thought (CoT) prompting with GPT-4, achieving a 33% relative improvement in the plan generation task [14].

LESSONS LEARNED

The study of World Models highlights a fundamental shift from reactive optimization toward predictive and model-based intelligence. World models enable agents to reason about future system evolution, yielding several key advantages for communication-oriented edge intelligence:

- **Multi-step Planning:** By internally simulating future trajectories, agents can evaluate long-term consequences [9]. This capability is particularly valuable in scenarios with temporally coupled decisions, such as adaptive beamforming, where anticipating user mobility and channel evolution enables proactive alignment and reduces reliance on frequent channel estimation.
- **Sample Efficiency:** World models learn the dynamics from limited real interactions and leverage imagined rollouts to refine policies [4]. This substantially reduces interaction overhead, which is critical in large-scale systems such as massive IoT networks, where exhaustive real-world exploration is impractical.
- **Handling Uncertainty:** Latent state representations in world models aggregate information over time, providing a structured memory for inferring partially observed [2]. This property supports robust decision-making under uncertainty, such as enabling predictive maintenance by inferring latent infrastructure degradation from noisy and partial network measurements.

The effectiveness of such predictive intelligence is further strengthened by emerging 6G technologies [4]. Semantic-aware communication enables the transmission of task-relevant latent representations, reducing communication overhead while preserving decision fidelity. In parallel, edge-cloud collaborative inference allows computationally demanding components of the world model to be offloaded to nearby infrastructure, while latency-critical operations remain on-device. In short, world models confer foresight, causal reasoning, and sample-efficient learning on edge devices, while 6G supplies the bandwidth, low latency, and semantic protocols to train, distribute, and maintain those models at scale.

WORLD MODEL FOR WIRELESS EDGE INTELLIGENCE OPTIMIZATION

PROPOSED FRAMEWORK

To realize these benefits, we propose Wireless Dreamer, a lightweight Q-learning framework that integrates a world model tailored for edge deployment. It is a Dreamer-style [1] framework tailored to solve sequential optimization problems

⁴ For additional implementation details, source codes, and experimental configurations, please refer to the tutorial page: https://changyuanzhao.github.io/Wireless_Dreamer.

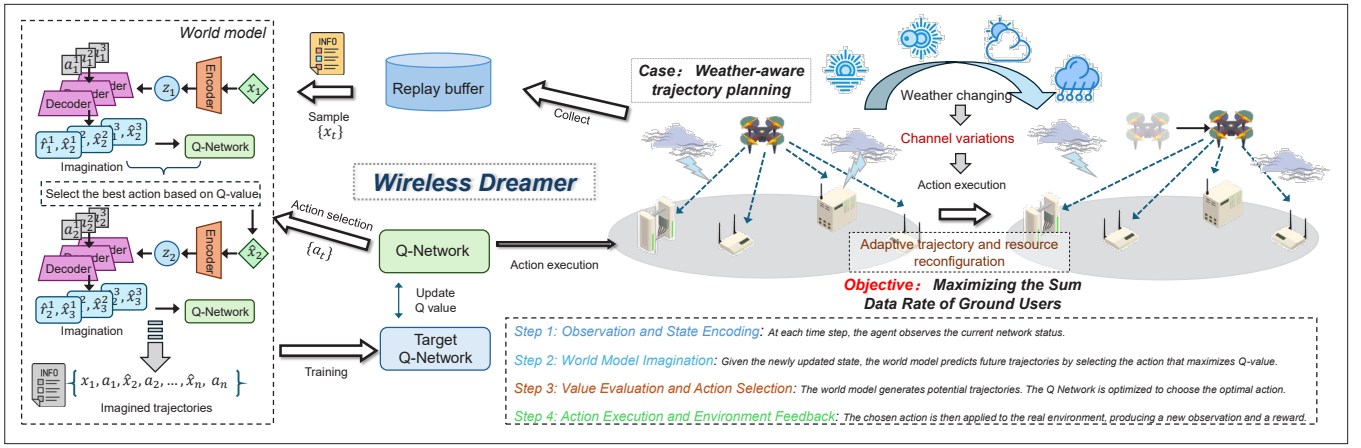


FIGURE 2. The workflow of the proposed framework. The left part is the model structure of Wireless Dreamer, including a world model, Q-network, and target Q-network. The bottom part presents the continuous processes of Wireless Dreamer. The right part illustrates the weather-aware trajectory planning in a UAV-assisted scenario in LAWNs.

in wireless edge networks with a discrete action space. As shown in Fig. 2 our framework consists of three main components:

- **World Model:** a learned dynamics model that predicts how the wireless network state changes over time. This model ingests the current observation, such as channel measurements and UAV locations, and produces a predicted next state and reward.
- **Q Network:** a neural network that estimates the long-term value of a given state. In Wireless Dreamer, the Q Network is trained on simulated trajectories from the world model.
- **Target Q Network:** a periodically updated replica of the primary Q Network, used to generate stable bootstrap targets during value learning.

WIRELESS DREAMER WORKFLOW OVERVIEW

To illustrate how the components interact, Fig. 2 and the following steps summarize the Wireless Dreamer workflow.⁴ The framework operates in a closed loop that iteratively performs observation, latent state update, imagination-based planning, action evaluation, and environment interaction, as detailed below:

- **Observation and Latent State Encoding:** At each time step t , the agent observes the current network state x_t and encodes it into a latent representation z_t .
- **Discrete Imagination over Action Space:** Given z_t , the world model enumerates all feasible discrete actions and generates the imagined trajectory for each action a_t .
- **Value Estimation and Action Evaluation:** The Q-network then evaluates each predicted state-action pair by estimating $Q(z_{t+1}, a_t)$, enabling direct comparison of long-term utility across candidate actions.
- **Action Execution and Environment Feedback:** The agent executes the action with the highest Q-value and receives the next observation x_{t+1} and reward r_t .

Throughout this workflow, Wireless Dreamer fully exploits the discrete nature of wireless control spaces by combining model-based prediction with Q-network evaluation, enabling action selection based on estimated cumulative return rather than instantaneous rewards, in contrast to conventional Dreamer-style policy sampling.

CASE STUDY: WEATHER-AWARE UAV TRAJECTORY PLANNING

SYSTEM MODEL

We consider a UAV-assisted wireless communication scenario in LAWNs, where a single UAV acts as a mobile base station to provide downlink coverage for multiple ground users under dynamic and spatially varying weather conditions, as shown in Fig. 2. The UAV traverses a two-dimensional area over a fixed time horizon and dynamically adjusts its position to optimize communication performance while responding to evolving environmental factors such as wind intensity.

This scenario involves one UAV and N users randomly distributed across a 64×64 grid-based area, where each grid is $4 \text{ m} \times 4 \text{ m}$. The UAV operates over a 28 GHz mmWave band with a total bandwidth of 100 MHz and a transmission power of 30 dBm. The weather is modeled by the Gaussian Field Model, which simulates a drifting Gaussian hotspot to represent dynamic environmental factors such as wind speed [15]. The hotspot's position evolves over time, directly affecting the communication path loss. The UAV aims to maximize the total downlink capacity across users within T time steps, while its trajectory is influenced by the need to avoid adverse weather zones and turbulence. This case study considers 10 users and a time horizon of 100 steps.

NUMERICAL RESULTS

Experimental Setup: Our experiments are conducted on a Linux server running Ubuntu 22.04 and equipped with an NVIDIA RTX A6000 GPU. We benchmark the proposed method against DQN [5], a random policy, and a Dreamer-style baseline that adopts a similar world-model architecture but without action evaluation. In addition, we evaluate the impact of different rollout horizons and imagination-related parameters, which respectively affect Q-network training through imagined trajectories and the future planning depth during action selection.

Performance Analysis: The average episodic rewards are shown in Fig. 3. Wireless Dreamer converges significantly faster than DQN, reaching stable performance around episode 350 compared with about 600 episodes for DQN,

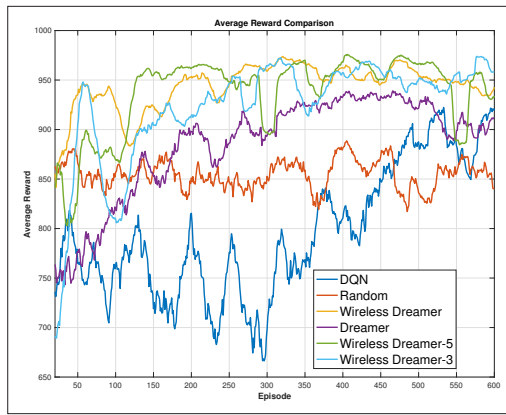


FIGURE 3. Comparison of average episodic rewards, where Wireless Dreamer achieves faster learning and superior policies compared to baseline methods.

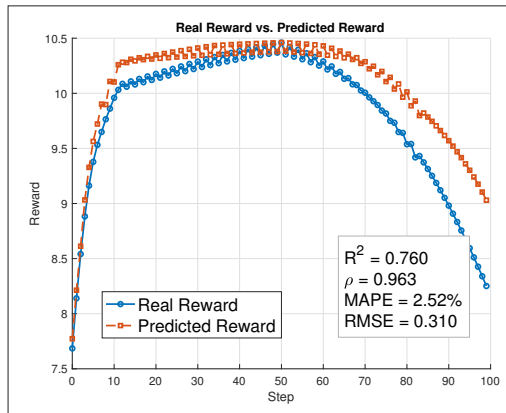


FIGURE 4. Comparison between real and predicted rewards, showing high consistency and low prediction error, particularly in the early stages.

corresponding to a convergence speedup of approximately 42%. This faster convergence indicates improved sample efficiency with fewer interactions with the environment.

During training, Dreamer-based methods incur higher computational costs due to world model learning and planning. However, during inference, the latency remains comparable to DQN, at about 0.01s over 100 steps, and even with planning enabled, the per-step decision time stays at the millisecond scale, indicating feasibility for real-time control in LAWNs.

We further examine the impact of rollout and planning horizons. Wireless Dreamer-3 employs rollout and planning horizons of 3, Wireless Dreamer-5 uses a rollout length of 5 with a planning horizon of 3, and the default Wireless Dreamer adopts rollout and planning horizons of 5. The results reveal a clear trade-off. Longer rollout horizons accelerate early-stage convergence by enhancing long-term reward estimation, but also introduce accumulated prediction errors that may affect stability. This observation suggests that moderate rollout and planning lengths provide a favorable balance between foresight and robustness in dynamic wireless environments.

Moreover, we evaluate the accuracy of the learned world model by comparing the predicted rewards with the real rewards during the evaluation phase. As shown in Fig. 4, the predicted rewards closely follow the real returns over the entire decision horizon, indicating a strong consistency

between model-based estimation and environment feedback. Quantitatively, the prediction achieves an R^2 score of 0.760 and a Pearson correlation coefficient of $\rho = 0.963$, demonstrating a high degree of linear agreement between the predicted and real rewards. In addition, the mean absolute percentage error (MAPE) is only 2.52%, with a root mean square error (RMSE) of 0.310, confirming the overall reliability of the reward prediction.

Key Takeaways from the Case Study: The results show that the proposed Wireless Dreamer improves sample efficiency by reducing interactions with the real environment and accelerating convergence, while learning key environmental dynamics that affect communication performance, such as UAV mobility and weather-induced channel variations. Moreover, the inference latency remains at the millisecond scale, indicating that the approach is feasible for real-time control in LAWNs.

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REFERENCES

- [1] Z. Zhu et al., "Is SORA A World Simulator? A Comprehensive Survey on General World Models and Beyond," arXiv preprint arXiv:2405.03520, 2024.
- [2] Y. Guan et al., "World Models for Autonomous Driving: An Initial Survey," *IEEE Trans. Intell. Transp. Syst.*, 2024.
- [3] J. Ding et al., "Understanding World or Predicting Future? A Comprehensive Survey of World Models," arXiv preprint arXiv:2411.14499, 2024.
- [4] C. Zhao et al., "Edge General Intelligence Through World Models and Agentic AI: Fundamentals, Solutions, and Challenges," arXiv preprint arXiv:2508.09561, 2025.
- [5] K. Li et al., "On-Board Deep Q-Network for UAV-Assisted Online Power Transfer and Data Collection," *IEEE Trans. Veh. Technol.*, vol. 68, no. 12, 2019, pp. 12,215–26.
- [6] L. U. Khan et al., "Digital Twin of Wireless Systems: Overview, Taxonomy, Challenges, and Opportunities," *IEEE Commun. Surv. Tut.*, vol. 24, no. 4, 2022, pp. 2230–54.
- [7] M. Xu et al., "A full Dive into Realizing the Edge-Enabled Metaverse: Visions, Enabling Technologies, and Challenges," *IEEE Commun. Surv. Tut.*, vol. 25, no. 1, 2022, pp. 656–700.
- [8] G. Qu et al., "Mobile Edge Intelligence for Large Language Models: A Contemporary Survey," *IEEE Commun. Surv. Tut.*, 2025.
- [9] D. Hafner et al., "Mastering Atari with Discrete World Models," arXiv preprint arXiv:2010.02193, 2020.
- [10] C. Zhao et al., "Generative AI for Secure Physical Layer Communications: A Survey," *IEEE Trans. Cogn. Commun. Netw.*, 2024.
- [11] C. Zhao et al., "Generative AI-Enabled Wireless Communications for Robust Low-Altitude Economy Networking," arXiv preprint arXiv:2502.18118, 2025.
- [12] D. Ha and J. Schmidhuber, "World Models," arXiv preprint arXiv:1803.10122, 2018.
- [13] X. Wang et al., "Worlddreamer: Towards General World Models for Video Generation via Predicting Masked Tokens," arXiv preprint arXiv:2401.09985, 2024.
- [14] S. Hao et al., "Reasoning with Language Model is Planning with World Model," arXiv preprint arXiv:2305.14992, 2023.
- [15] N. Chen et al., "Wind Power Forecasts Using Gaussian Processes and Numerical Weather Prediction," *IEEE Trans. Power Syst.*, vol. 29, no. 2, 2013, pp. 656–65.

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